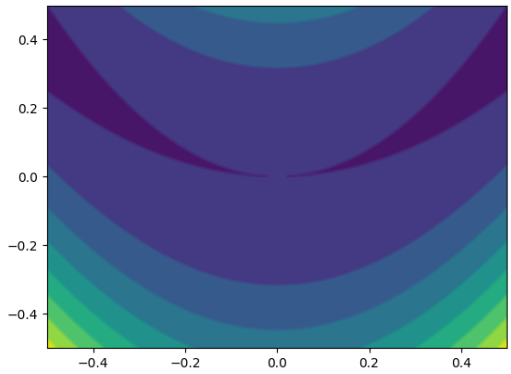
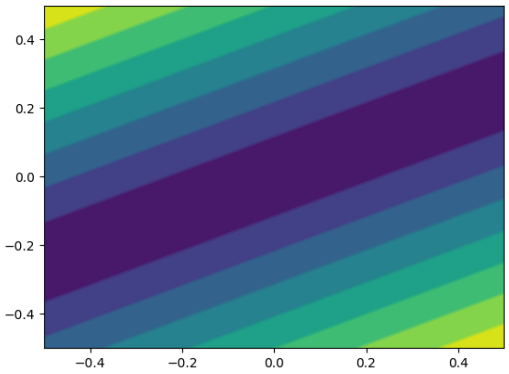
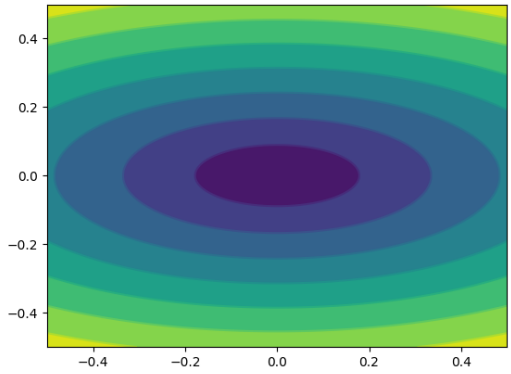
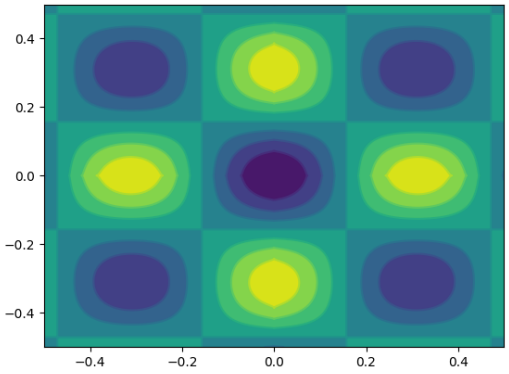
SHEET 2 – **OPTIMIZATION FOR DATA SCIENCE** SS 2026
DATE: 09.06.2026

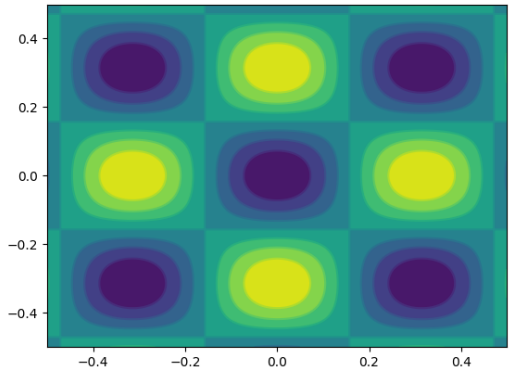
Discussion: Given the following contour plots of functions $f : \mathbb{R}^2 \to \mathbb{R}$. Discuss which of the following functions are convex and the location of local, global, strict local and strict global minima. See also the accompanying Jupyter notebook.











Exercise 2.1

Let $n \in \mathbb{N}$ and $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable. Let $\epsilon > 0$ and $x, d \in \mathbb{R}^n$. Taylor series expansion with Lagrangian remainder yields that there exists $\theta_\epsilon \in [0, \epsilon]$

$$f(x + \epsilon d) = f(x) + \epsilon \nabla f(x)^\top d + \frac{1}{2}\epsilon^2 d^\top \nabla^2 f(x + \theta_\epsilon d) d.$$

Let $(\epsilon_n)_{n\in\mathbb{N}}$ be a strictly monotonically decreasing sequence with $\lim_{n\to\infty} \epsilon_n = 0$ and such that $1 < c \leq \frac{\epsilon_{n-1}}{\epsilon_n}$. Verify that if $d^\top \nabla^2 f(x) d \neq 0$ we have

$$\lim_{n\to\infty} \frac{\ln\left(\left|\frac{f(x+\epsilon_{n-1}d) - f(x) - \epsilon_{n-1}\nabla f(x)^\top d}{f(x+\epsilon_n d) - f(x) - \epsilon_n \nabla f(x)^\top d}\right|\right)}{\ln\left(\frac{\epsilon_{n-1}}{\epsilon_n}\right)} = 2. \tag{1}$$

Use this formula to implement a function `taylor_test` that serves as an indicator if the implementation of the derivative is correct. It takes as input

- a function handle for an arbitrary function `f`,
- a function handle for its derivative `df`,
- a point `x`,
- a direction `d`,
- a finite list `epslist` of strictly monotonically decreasing ϵ,

and raises an error if the quotient given in (1) does not approximate 2.

Exercise 2.2

Let $f : \mathbb{R}^n \to \mathbb{R}$ be strongly convex with modulus $\mu > 0$. Show that for $x, y \in \mathbb{R}^n$ we have

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|y - x\|_2^2.$$

Exercise 2.3

Let the function $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x, y) = y^2 - 3yx^2 + 2x^4. \tag{2}$$

1. Compute all stationary points of f.
2. Prove that in $(0, 0)$ the function is non-decreasing in each direction $d \in \mathbb{R}^2$: Let $\alpha_d(t) := td$ with $d \in \mathbb{R}^2 \setminus \{0\}$. Show that the function $\Phi_d(t) := f(\alpha_d(t))$ is bounded from below and has a strict local minimum in $t = 0$.
3. Is $(0, 0)$ a local minimum of f?

Exercise 2.4

Let $n \in \mathbb{N}$.

1. Prove that there is no function $f : \mathbb{R}^n \to \mathbb{R}$, which is strongly convex and Lipschitz continuous.
2. Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is convex and upper-bounded. Show that f must be a constant function.